MIL-S-19500/37D
22 November 1971
SUPERSEDING
MIL-S-19500/37C
9 December 1964
(See 6.3)

MILITARY SPECIFICATION

SEMICONDUCTOR DEVICE, TRANSISTOR, NPN, SILICON, LOW-POWER TYPES 2N333, 2N335, 2N336, 2N333A, 2N335A, AND 2N336A

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for silicon, NPN, low-power transistors.

1.2 Physical dimensions. See figure 1 (TO-5).

1.3 Maximum ratings.

P_T T_A = 25°C		V_{CEO}	V_{CBO}	V_{EBO}		T_{stg}	
2N333 2N335 2N336	2N333A 2N335A 2N336A			2N333 2N335 2N336	2N333A 2N335A 2N336A	2N333 2N335 2N336	2N333A 2N335A 2N336A
mW	mW	Vdc	Vdc	Vdc	Vdc	°C	°C
150 1/	500 2/	45	45	1	4	-65 to +175	-65 to +200

1/ Derate approximately 1 mW/°C for T_A between 25°C and 175°C.
2/ Derate approximately 3.33 mW/°C for T_A between 25°C and 175°C.

1.4 Primary electrical characteristics.

Limits	h_{fe} V_{CB} = 5 Vdc, I_E = -1 mAdc			$|h_{fb}|$ V_{CB} = 5 Vdc I_E = -1 mAdc	C_{obo} V_{CB} = 5 Vdc I_E = 0 100 kHz ≤ f ≤ 1 MHz
	2N333 2N333A	2N335 2N335A	2N336 2N336A		
				MHz	pF
Min	18	37	76	2.5	---
Max	44	90	270	---	15

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-S-19500 - Semiconductor Devices, General Specification for.

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STANDARDS

MILITARY

MIL-STD-202 - Test Methods for Electronic and Electrical Component Parts.
MIL-STD-750 - Test Methods for Semiconductor Devices.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General. Requirements shall be in accordance with MIL-S-19500, and as specified herein.

3.2 Abbreviations, symbols, and definitions. The abbreviations, symbols, and definitions used herein are defined in MIL-S-19500.

3.3 Design, construction, and physical dimensions. Transistors shall be of the design, construction, and physical dimensions shown on figure 1.

3.3.1 Lead material and finish. Lead material shall be Kovar or Alloy 52. Lead finish shall be gold-plated. (Leads may be tin-coated if specified in the contract or order, and this requirement shall not be construed as adversely affecting the qualified-product status of the device, or applicable JAN marking, see 6.2).

3.3.1.1 Selectivity of lead material. Where choice of lead material (see 3.3.1) is desired, it shall be specified in the contract or order (see 6.2).

3.3.2 Terminal-lead length. Terminal-lead length(s) other than that specified in figure 1 may be furnished when so stipulated under contract or order (see 6.2) where the devices covered herein are required directly for particular equipment-circuit installation or for automatic-assembly-technique programs. Where other lead lengths are required and provided, it shall not be construed as adversely affecting the qualified-product status of the device, or applicable JAN marking.

3.4 Performance characteristics. Performance characteristics shall be as specified in tables I, II, and III.

3.5 Marking. The following marking specified in MIL-S-19500 may be omitted from the body of the transistor at the option of the manufacturer:

(a) Country of origin.
(b) Manufacturer's identification.

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection shall be in accordance with MIL-S-19500, and as specified herein.

4.2 Qualification inspection. Qualification inspection shall consist of the examinations and tests specified in tables I, II, and III.

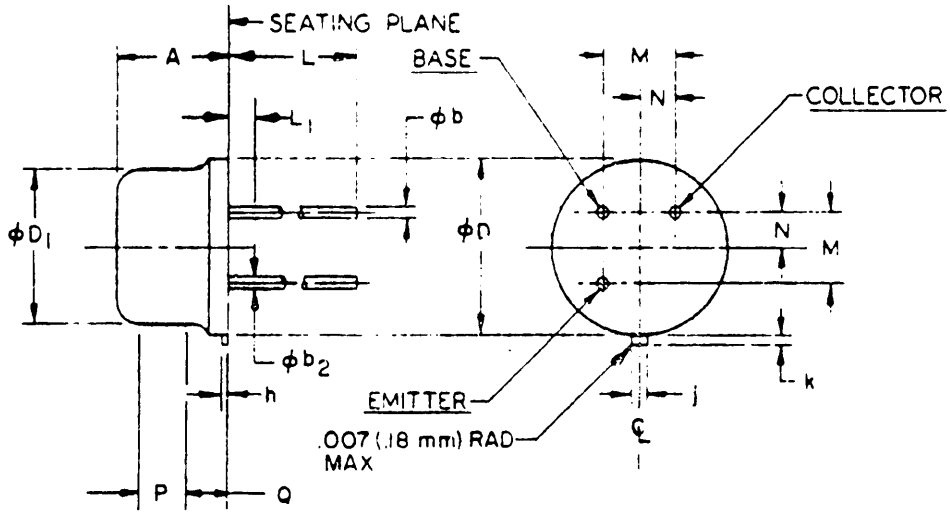
4.3 Quality conformance inspection. Quality conformance inspection shall consist of groups A, B, and C inspections.

4.3.1 Group A inspection. Group A inspection shall consist of the examinations and tests specified in table I.

4.3.2 Group B inspection. Group B inspection shall consist of the examinations and tests specified in table II.

4.3.3 Group C inspection. Group C inspection shall consist of the examinations and tests specified in table III. This inspection shall be conducted on the initial lot and thereafter every 6 months during production.

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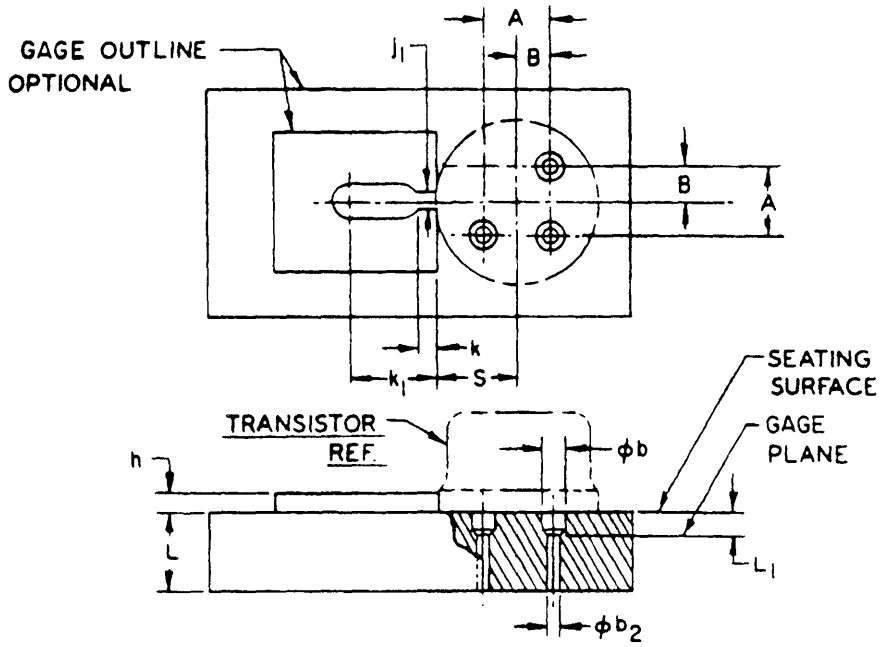


LTR	DIMENSIONS				NOTES
	INCHES		MILLIMETERS		
	MIN	MAX	MIN	MAX	
ØD	.335	.370	8.51	9.40	
$ØD_1$	.305	.335	7.75	8.51	
A	.240	.260	6.10	6.60	
L	1.500	1.750	38.10	44.45	9
L_1	---	.050	---	1.27	10
Øb	.016	.021	.41	.53	2, 9
$Øb_2$	.016	.019	.41	.48	3, 9
P	.100	---	2.54	---	4
Q	---	---	---	---	5
k	.029	.045	.74	1.14	8
j	.028	.034	.71	.86	
h	.009	.041	.23	1.04	
M	.1414 Nom		3.59 Nom		6
N	.0707 Nom		1.80 Nom		6

NOTES:
1. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
2. Measured in the zone beyond .250(6.35 mm) from the seating plane.
3. Measured in the zone .050(1.27 mm) and .250(6.35 mm) from the seating plane.
4. Variations on dimension $ØD_1$ in this zone shall not exceed .010(.25 mm).
5. Outline in this zone is not controlled.
6. When measured in a gaging plane .054+.001, -.000(1.37+.03, -.00 mm) below the seating plane of the transistor, maximum diameter leads shall be within .007(.18 mm) of their true location relative to a maximum width tab. Smaller diameter leads shall fall within the outline of the maximum diameter lead tolerance. Figure 2 shows the preferred measured method.
7. All leads electrically isolated from the case.
8. Measured from the maximum diameter of the actual device.
9. All 3 leads (see 3.3.1 and 3.3.2).
10. Diameter of leads in this zone is not controlled.

FIGURE 1. Physical dimensions of transistors, types 2N333, 2N333A, 2N335, 2N335A, 2N336, and 2N336A (TO-5).

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LTR	DIMENSIONS				NOTES
	INCHES		MILLIMETERS		
	MIN	MAX	MIN	MAX	
A	.1409	.1419	3.58	3.60	
B	.0702	.0712	1.78	1.81	
S	.182	.199	4.62	5.05	
k	.009	.011	.23	.28	
k_1	.125 Nom		3.18 Nom		
L_1	.054	.055	1.37	1.40	
L	.372	.378	9.45	9.60	
j_1	.0350	.0355	.89	.90	
h	.150 Nom		3.81 Nom		
ϕb_2	.0325	.0335	.83	.85	
ϕb	.0595	.0605	1.51	1.54	

NOTES:
1. The following gaging procedure shall be used: The use of a pin straightener prior to insertion in the gage is permissible. The device being measured shall be inserted until its seating plane is .125±.010(3.18±.25mm) from the seating surface of the gage. A spacer may be used to obtain the .125(3.18 mm) distance from the gage seat prior to force application. A force of 8 oz ± .50 oz shall then be applied parallel and symmetrical to the device's cylindrical axis. When examined visually after the force application (the force need not be removed) the seating plane of the device shall be seated against the gage.
2. The location of the tab locator, within the limits of dimension S, will be determined by the tab and flange dimension of the device being checked.
3. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.

FIGURE 2. Gage for lead and tab location for transistors, types 2N333, 2N333A, 2N335, 2N335A, 2N336, and 2N336A.

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TABLE I. Group A inspection.

Examination or test	MIL-STD-750		LTPD	Symbol	Limits		Unit
	Method	Details			Min	Max	
Subgroup 1			10				
Visual and mechanical examination	2071				---	---	---
Subgroup 2			5				
Collector to emitter cutoff current	3041	Bias cond. D; V_{CE} = 30 Vdc		I_{CEO}	---	1.0	µAdc
Collector to base cutoff current	3036	Bias cond. D; V_{CB} = 30 Vdc		I_{CBO}	---	0.5	µAdc
Breakdown voltage, collector to base	3001	Bias cond. D; I_C = 50 µAdc		BV_{CBO}	45	---	Vdc
Breakdown voltage, collector to emitter	3011	Bias cond. D; I_C = 1 mAdc		BV_{CEO}	45	---	Vdc
Breakdown voltage, collector to emitter	3011	Bias cond. D; I_C = 100 µAdc		BV_{CEO}	45	---	Vdc
Breakdown voltage, emitter to base	3026	Bias cond. D		BV_{EBO}			
2N333, 2N335, 2N336		I_E = 10 µAdc			1.0	---	Vdc
2N333A, 2N335A, 2N336A		I_E = 100 µAdc			4.0	---	Vdc
Collector to emitter voltage (saturated)	3071	I_C = 5 mAdc		$V_{CE(sat)}$			
2N333, 2N335, 2N336		I_B = 2.2 mAdc			---	1.0	Vdc
2N333A, 2N335A, 2N336A		I_B = 1.0 mAdc			---	1.0	Vdc
Subgroup 3			5				
Small-signal open-circuit output admittance	3216	V_{CB} = 5 Vdc; I_E = -1 mAdc		h_{ob}	0	1.2	µmho
Small-signal open-circuit reverse-voltage transfer ratio	3211	V_{CB} = 5 Vdc; I_E = -1 mAdc		h_{rb}	0	$1x10^{-3}$	---
Small-signal short-circuit input impedance	3201	V_{CB} = 5 Vdc; I_E = -1 mAdc		h_{ib}	30	80	ohms
Small-signal short-circuit forward-current transfer ratio	3206	V_{CB} = 5 Vdc; I_E = -1 mAdc		h_{fe}			
2N333, 2N333A					18	44	---
2N335, 2N335A					37	90	---
2N336, 2N336A					76	270	---
Forward-current transfer ratio	3076	V_{CE} = 5 Vdc; I_C = 10 mAdc		h_{FE}			
2N333, 2N333A					6	60	---
2N335, 2N335A					12	120	---
2N336, 2N336A					25	275	---

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TABLE I. Group A inspection. - Continued

Examination or test	MIL-STD-750		LTPD	Symbol	Limits		Unit
	Method	Details			Min	Max	
Subgroup 3 - Continued							
Forward-current transfer ratio	3076	V_{CE} = 5 Vdc; I_C = 100 μAdc		h_{FE}			
2N333, 2N333A					3	---	---
2N335, 2N335A					6	---	---
2N336, 2N336A					15	---	---
Subgroup 4			10				
Magnitude of common-emitter small-signal short-circuit forward-current transfer ratio	3306	V_{CE} = 5 Vdc; I_C = 1 mAdc; f = 2.5 MHz		$\lvert h_{fe} \rvert$			
2N333					---	40	---
2N335					---	60	---
2N336					---	120	---
2N333A					---	30	---
2N335A					---	30	---
2N336A					---	30	---
Magnitude of common-emitter small-signal short-circuit forward-current transfer ratio	3306	V_{CE} = 5 Vdc; I_C = 1 mAdc; f = 10 MHz		$\lvert h_{fe} \rvert$			
2N333					---	13	---
2N335					---	15	---
2N336					---	17	---
2N333A					---	13	---
2N335A					---	13	---
2N336A					---	13	---
Small-signal short-circuit forward-current transfer-ratio, cutoff frequency	3301	V_{CB} = 5 Vdc; I_E = -1 mAdc		f_{hfb}	2.5	---	MHz
Subgroup 5			10				
Open circuit output capacitance	3236	V_{CB} = 5 Vdc; V_{sig} = 0.1 Vac; 100 kHz $\leq f \leq$ 1 MHz		C_{obo}	---	15	pF
Noise figure	3246	V_{CB} = 5 Vdc; I_E = -1 mAdc; f = 1 kHz; R_g = 500 ohms		NF	---	30	dB
Subgroup 6			10				
High-temperature operation:		T_A = +150°C					
Collector to base cutoff current	3036	Bias cond. D; V_{CB} = 30 Vdc		I_{CBO}	---	25	μAdc
Low-temperature operation:		T_A = -65°C					
Small-signal short-circuit forward-current transfer ratio	3206	V_{CB} = 5 Vdc; I_E = -1 mAdc		h_{fe}			
2N333, 2N333A					8	---	---
2N335, 2N335A					18	---	---
2N336, 2N336A					35	---	---

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TABLE II. Group B inspection.

Examination or test	MIL-STD-750		LTPD	Symbol	Limits		Unit
	Method	Details			Min	Max	
Subgroup 1			20				
Physical dimensions	2066	(See figure 1)			---	---	---
Subgroup 2			15				
Solderability	2026			---	---	---	---
Thermal shock (temperature cycling)	1051	Test cond. C, except: 10 cycles; exposure time at temp. extremes = 15 minutes (min); and in step 3. T_{max} = 175°C		---	---	---	---
Thermal shock (glass strain)	1056	Test cond. A		---	---	---	---
Moisture resistance	1021			---	---	---	---
End points:							
Collector to base cutoff current	3036	Bias cond. D; V_{CB} = 30 Vdc		I_{CBO}	---	0.5	μAdc
Small-signal short-circuit forward-current transfer ratio	3206	V_{CB} = 5 Vdc; I_E = -1 mAdc		h_{fe}			
2N333, 2N333A					18	44	---
2N335, 2N335A					37	90	---
2N336, 2N336A					76	270	---
Subgroup 3			15				
Shock (2N333, 2N335, 2N336)	2016	Nonoperating; 500 G; 1 ms; 5 blows in each orientation X_1, Y_1, Y_2, and Z_1		---	---	---	---
Shock (2N333A, 2N335A, 2N336A)	2016	Nonoperating; 1500 G; approx. 0.5 ms; 5 blows in each orientation X_1, Y_1, Y_2, and Z_1		---	---	---	---
Vibration, variable frequency	2056	Nonoperating		---	---	---	---
Constant acceleration	2006	20,000 G in each orientation X_1, Y_1, Y_2, and Z_1		---	---	---	---
End points: (Same as subgroup 2)							
Subgroup 4			15				
Terminal strength (lead fatigue)	2036	Test cond. E		---	---	---	---
End points: Hermetic seal	1071	Test cond. G or H for fine leaks; test cond. A, C, D, or F for gross leaks		---	---	$1x10^{-7}$	atm cc/sec

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TABLE II. Group B inspection. - Continued

Examination or test	MIL-STD-750		LTPD	Symbol	Limits		Unit
	Method	Details			Min	Max	
Subgroup 5			15				
Salt atmosphere (corrosion)	1041			---	---	---	---
Subgroup 6			7				
High temperature life (nonoperating)	1032	Time = 340 hours (see 4.3.4)		---	---	---	---
2N333, 2N335, 2N336 2N333A, 2N335A, 2N336A		T_{stg} = 175°C T_{stg} = 200°C					
End points:							
Collector to base cutoff current	3036	Bias cond. D; V_{CB} = 30 Vdc		I_{CBO}	---	1.0	μAdc
Breakdown voltage, collector to base	3001	Bias cond. D; I_C = 50 μAdc		BV_{CBO}	40	---	Vdc
Forward-current transfer ratio	3076	V_{CE} = 5 Vdc; I_C = 10 mAdc		Δh_{FE}	---	±25	% change from initial recorded value
Small-signal short-circuit forward-current transfer ratio	3206	V_{CB} = 5 Vdc; I_E = -1 mAdc		h_{fe}			
2N333, 2N333A					15	---	---
2N335, 2N335A					30	---	---
2N336, 2N336A					60	---	---
Subgroup 7			7				
Steady-state operation life (2N333, 2N335, 2N336)	1027	T_A = 125°C; V_{CB} = 20 Vdc; P_T = 50 mW; time = 340 hours (see 4.3.4)		---	---	---	---
End points: (Same as subgroup 6)							
Steady-state operation life (2N333A, 2N335A, 2N336A)	1027	T_A = 25°C; V_{CB} = 30 Vdc; P_T = 500 mW; time = 340 hours (see 4.3.4)		---	---	---	---
End points: (Same as subgroup 6)							

TABLE III. Group C inspection.

Examination or test	MIL-STD-750		LTPD	Symbol	Limits		Unit
	Method	Details			Min	Max	
Subgroup 1			10				
Resistance to solvents	---	MIL-STD-202, Method 215 (see 4.4.1)		---	---	---	---

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TABLE III. Group C inspection. - Continued

Examination or test	MIL-STD-750		LTPD	Symbol	Limits		Unit
	Method	Details			Min	Max	
Subgroup 2			$\lambda = 10$				
High-temperature life (nonoperating)	1031	(see 4.3.4)		---	---	---	---
2N333, 2N335, 2N336		T_{stg} = 175°C					
2N333A, 2N335A, 2N336A		T_{stg} = 200°C					
End points: (Same as subgroup 6 of group B)							
Subgroup 3			$\lambda = 10$				
Steady-state operation life (2N333, 2N335, 2N336)	1026	T_A = 125°C; V_{CB} = 20 Vdc; P_T = 50 mW (see 4.3.4)		---	---	---	---
End points: (Same as subgroup 6 of group B)							
Steady-state operation life (2N333A, 2N335A, 2N336A)	1026	T_A = 25°C; V_{CB} = 30 Vdc; P_T = 500 mW (see 4.3.4)		---	---	---	---
End points: (Same as subgroup 6 of group B)							

4.3.4 Groups B and C life-test samples. Sample units that have been subjected to the group B 340-hour life test may be continued on test to 1,000 hours in order to satisfy group C life-test requirements. These sample units shall be predesignated, and shall remain subjected to the group C 1,000-hour acceptance evaluation after they have passed the group B 340-hour acceptance criteria. The cumulative total of failures found during the 340-hour test and during the subsequent interval up to 1,000 hours shall be computed for the 1,000-hour acceptance criteria (see 4.3.3).

4.3.5 Representative lot (groups B and C inspections). At the option of the manufacturer, any of the types 2N333, 2N335, and 2N336 may be used for groups B and C inspections as representative of a lot containing the prototypes, provided the type so selected is in the lot. Any of the types 2N333A, 2N335A, and 2N336A may be used for groups B and C inspections as representative of a lot containing the "A" types, provided the type so selected is in the lot.

4.4 Methods of examination and test. Methods of examination and test shall be as specified in tables I, II, and III, and as follows:

4.4.1 Resistance to solvents. Transistors shall be tested in accordance with method 215 of MIL-STD-202. The following details shall apply:

(a) All areas of the transistor body where marking has been applied shall be brushed.
(b) After subjection to the tests, there shall be no evidence of mechanical damage to the device and markings shall have remained legible.

5. PREPARATION FOR DELIVERY

5.1 See MIL-S-19500, section 5.

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6. NOTES

6.1 Notes. The notes specified in MIL-S-19500 are applicable to this specification.

6.2 Ordering data.

(a) Lead finish if other than gold-plated (see 3.3.1).
(b) Lead material (see 3.3.1.1)
(c) Terminal-lead length if other than as specified in figure 1 (see 3.3.2).

6.3 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodians:
Army - EL
Navy - EC
Air Force - 17

Review activities:
Army - MU, MI
Navy -
Air Force - 11, 70, 80
DSA - ES

User activities:
Army - SM
Navy - CG, MC, SH, AS
Air Force -

Preparing activity:
Navy - EC

Agent:
DSA - ES

(Project 5961-0215)

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